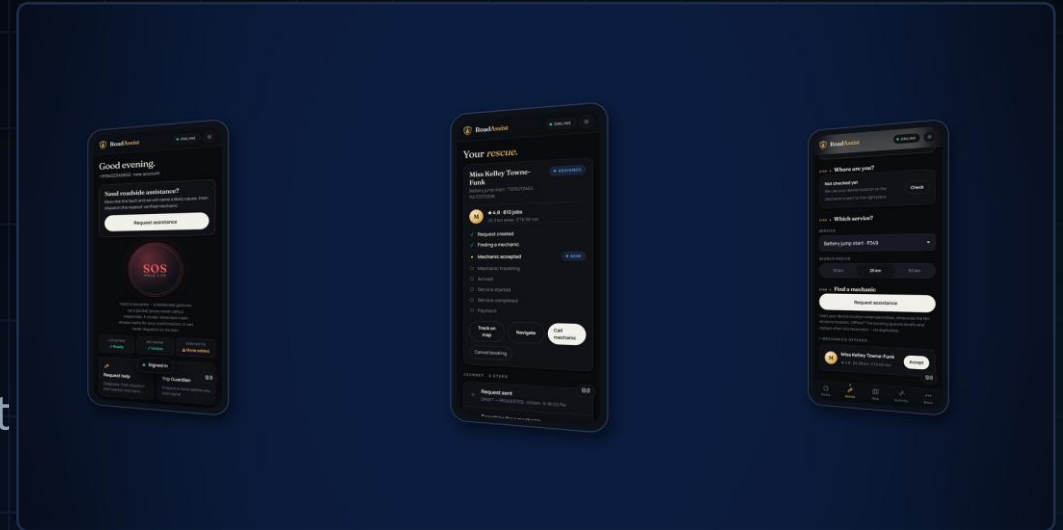


SWE4004 · CLOUD COMPUTING AND APPLICATIONS

RoadAssist Bharat

AI-Powered · Cloud-Connected · Network-Resilient
Emergency Mobility Platform

"RoadAssist doesn't stop when the network stops."



TEAM

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INSTITUTION

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THE PROBLEM

A breakdown is not a software problem

...until you realise the app you would use to fix it needs the network you do not have.

21:40

A Swift stops on NH-48. Two hundred kilometres from home, one bar of signal.

21:41

The driver opens a roadside-assistance app. It shows a spinner.

21:44

Still a spinner. There is no way to tell whether anything was sent.

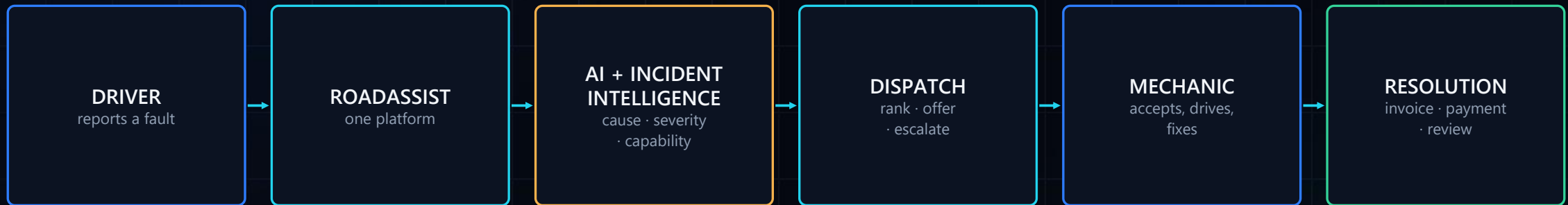
21:52

They give up and start calling numbers from a sticker on the windscreen.

WHAT ACTUALLY FAILS

- Coverage is worst exactly where breakdowns are most dangerous
- No way to know if a request was sent
- No centralised incident record
- Response time is unmeasured, so it cannot be improved

One platform, from incident to resolution



Every step is real application state. No manual database editing anywhere in the normal workflow.

Six things, and one that matters most

AI-ASSISTED DIAGNOSIS

Rules engine, explainable,
same table on device and cloud

DYNAMIC DISPATCH

Ranked, wave-based,
busy providers excluded

REAL-TIME TRACKING

Server-sent events,
measured at 65 ms

SECURE PAYMENTS

Server-side amount,
signature-verified webhook

INCIDENT INTELLIGENCE

Severity, capability,
tamper-evident audit

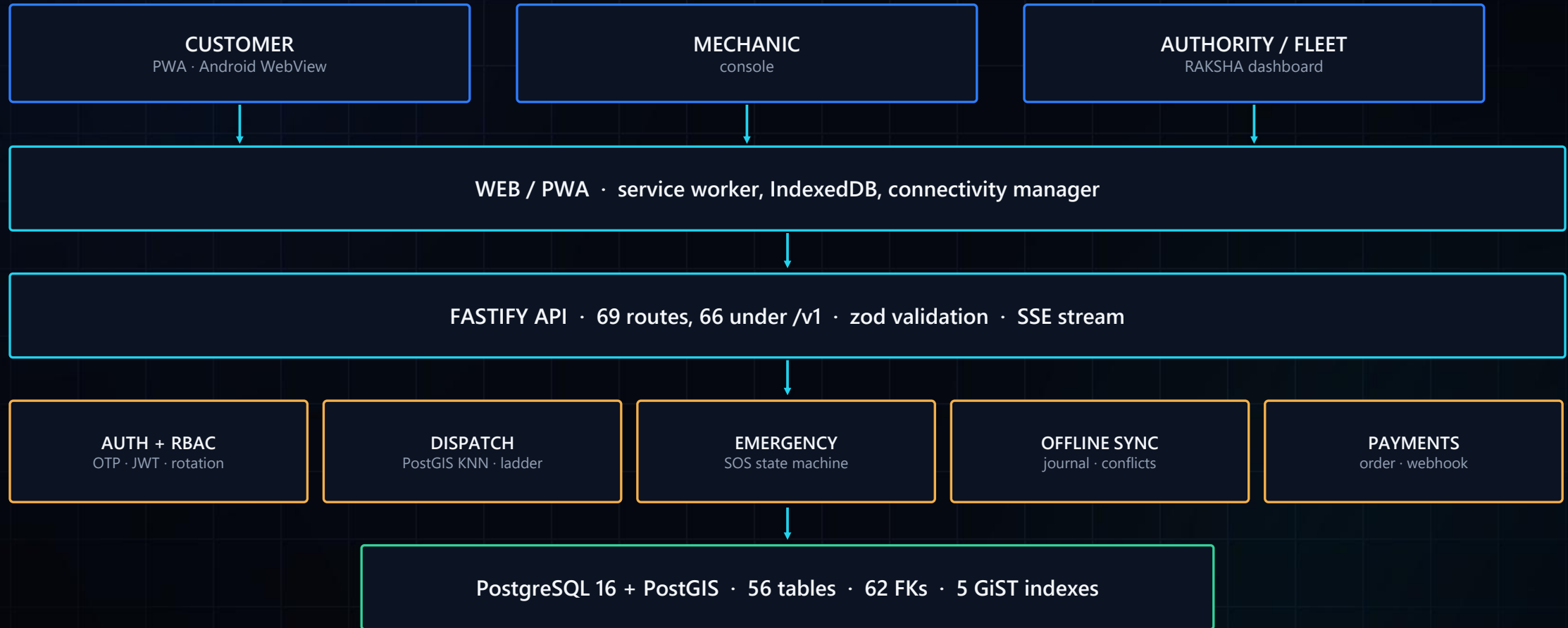
OFFLINE-FIRST EMERGENCY

SOS with no network,
stored and forwarded

“RoadAssist doesn't stop when the network stops.”

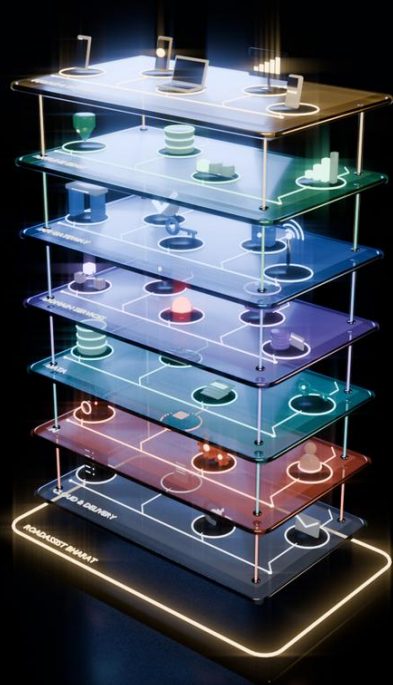
System architecture

One container, one PostGIS database — live on Render + Neon in Singapore. Everything external is behind an adapter.



NEW · MODELLED IN BLENDER · DRAWN WITH THREE.JS · FED BY THE RUNNING PLATFORM

Every layer, live in 3D



app.roadassistbharat.online/layers.html

IMPLEMENTED

- Modelled in Blender 5.2 by a script: 7 layers, 31 parts, each a model of what it stands for; Draco-compressed to 595 KB
- Drawn in the browser with three.js: bloom, reflections, turn it any way, zoom, arrow keys
- Live every 8 s: API round trip, database latency, detections and corridor health
- Three journeys, step by step: a booking, an SOS with no signal, RAKSHA finding a pothole
- Embedded on the project site, the app home and the showcase; opens inside the Android app
- Honest labels: the moving packets are illustration; detection positions are simulated

Two paths, and the second one is the product



The connectivity manager decides which path runs — ONLINE, LIMITED or OFF-GRID — from measured evidence, never from navigator.onLine alone.

Off-grid SOS

No internet does not mean the emergency workflow disappears. It means it runs here instead.



WHAT THE USER IS TOLD

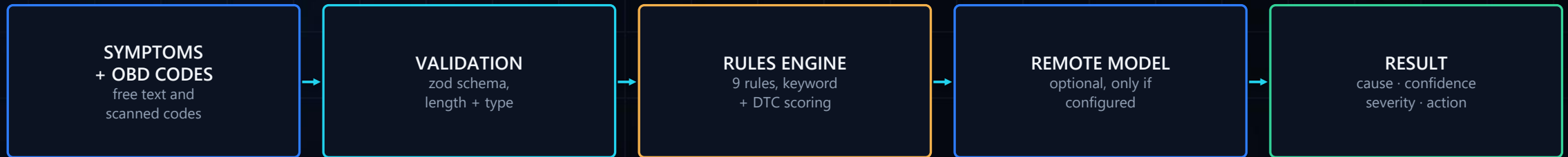
OFF-GRID SOS CREATED

No network connection detected.

Your emergency information is stored securely on this device and will be synchronized automatically when connectivity returns.

"Nothing has been transmitted."
We never claim emergency services were contacted. The 112 handoff is a stub and the API says so.

Diagnosis: rules first, model optional



THE SAFETY ASYMMETRY

A model may make a drivability verdict STRICTER. It can never make one laxer.

The rules result is always computed. A remote answer is used only if it clears AI_MIN_CONFIDENCE and does not contradict the rules on safety.

STATED PLAINLY

This is a deterministic rules engine, not machine learning. The UI labels it "rules-1.0.0", and "LOCAL OFFLINE DIAGNOSIS" on the device.

Trained models DO exist — YOLO11 for road damage: best run YOLO11s, mAP50 0.472; RAKSHA's demo detections come from a YOLO11n, mAP50 0.443. Both measured.

EXPLAINABLE, NOT OPAQUE

Incident intelligence

Every field below is derived from a rule you can read, not from a weighting nobody can inspect.



EXPLAINABLE

“Battery discharged or terminals loose” — because these keywords matched, at this confidence

AUDITABLE

Every prediction writes a model_predictions row: capability, version, confidence, latency, input HASH (never the text)

HONEST

Below 55% confidence the UI says so: “At 42% this is a suggestion, not a finding”

NEVER FABRICATED

No ETA unless a provider is travelling. No location unless GPS returned one.

Dispatch is dynamic scheduling

Work assigned at run time from a pool, by a score computed from live state, with timeout-driven re-scheduling.



THE BUG THIS AUDIT FOUND

Dispatch filtered on `is_available` alone — a duty toggle meaning “I am working today”, not “I am free now”. A mechanic already driving to one breakdown kept receiving offers for the next.

Fixed: provider state is DERIVED from live data.

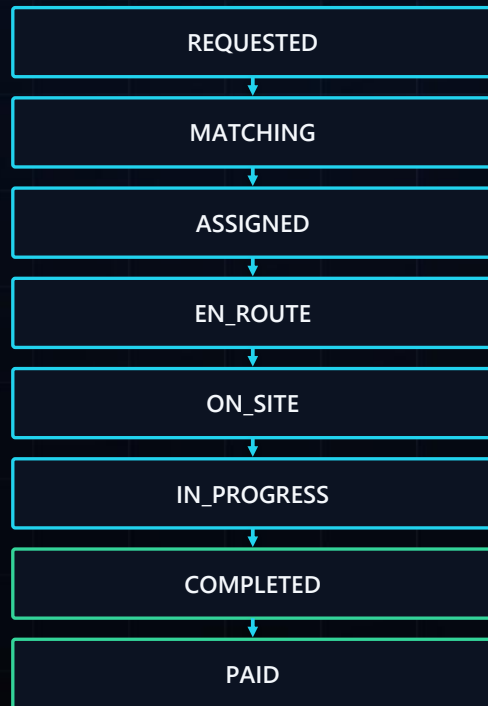
PROVIDER STATES

OFFLINE	off duty — never offered
AVAILABLE	free — offerable
OFFERED	holding an offer — still offerable
BUSY / EN_ROUTE / ON_JOB	committed — never offered

Two guarded state machines

A client sends a command; only the transition table decides whether it is legal. Every change writes a timestamped event row.

BOOKING — 13 states, 15 commands **EMERGENCY — ADR-0005 in code**



THE RULE THAT IS ENFORCED

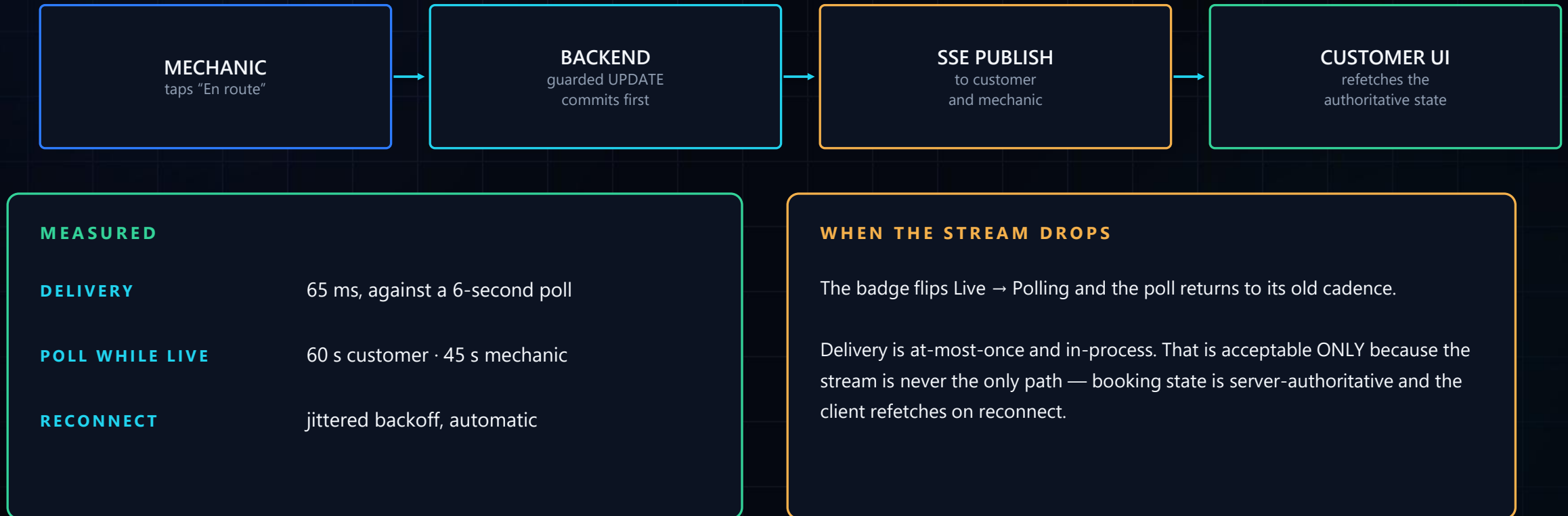
A model raises a SIGNAL. A human confirms an INCIDENT.

“escalate” is reachable only from CONFIRMED. There is no path from a model-detected crash to a dispatched responder that does not pass through a person.

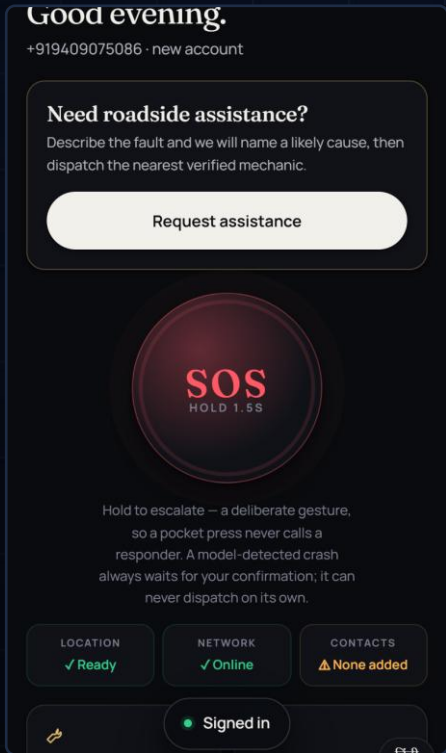
That is one line in a transition table, and it is unit-tested.

Real-time, but never the source of truth

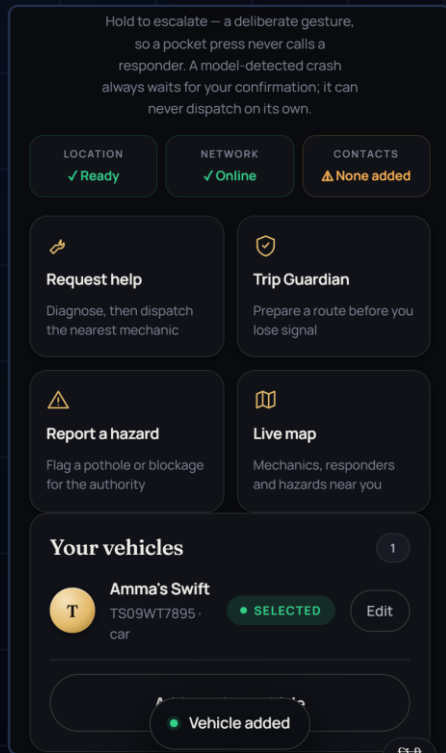
Server-sent events over one long-lived GET. Persist first, publish second — always.



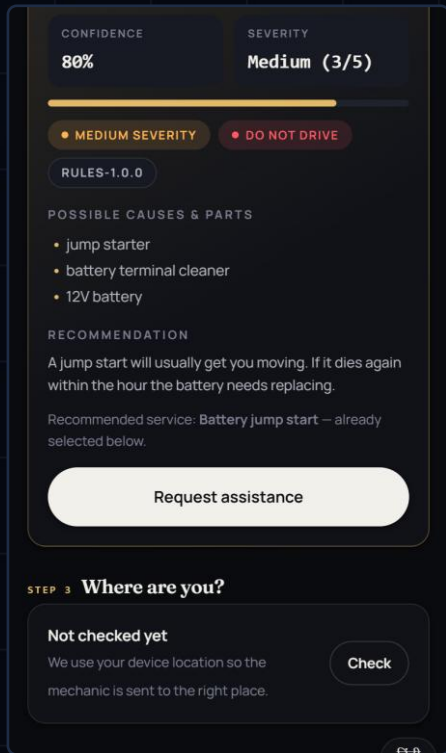
Customer application



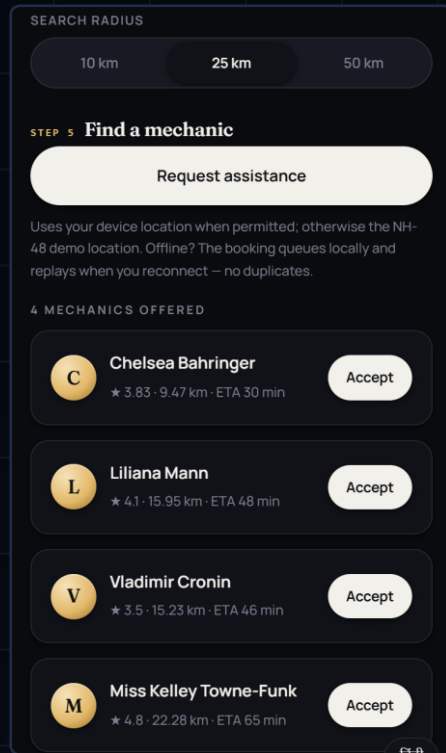
Home · SOS + readiness



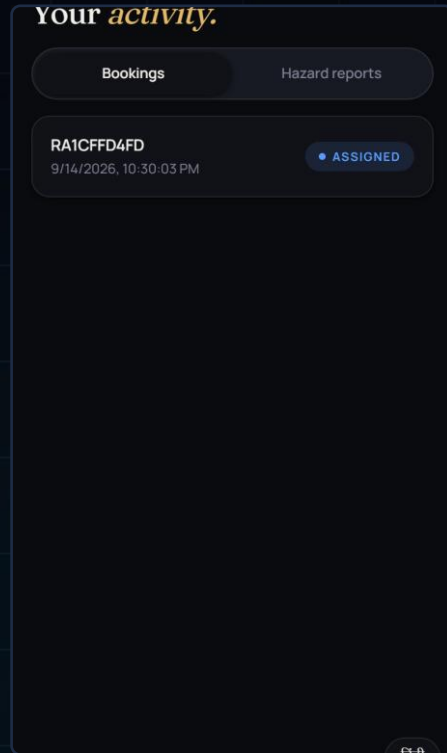
Vehicle



Diagnosis



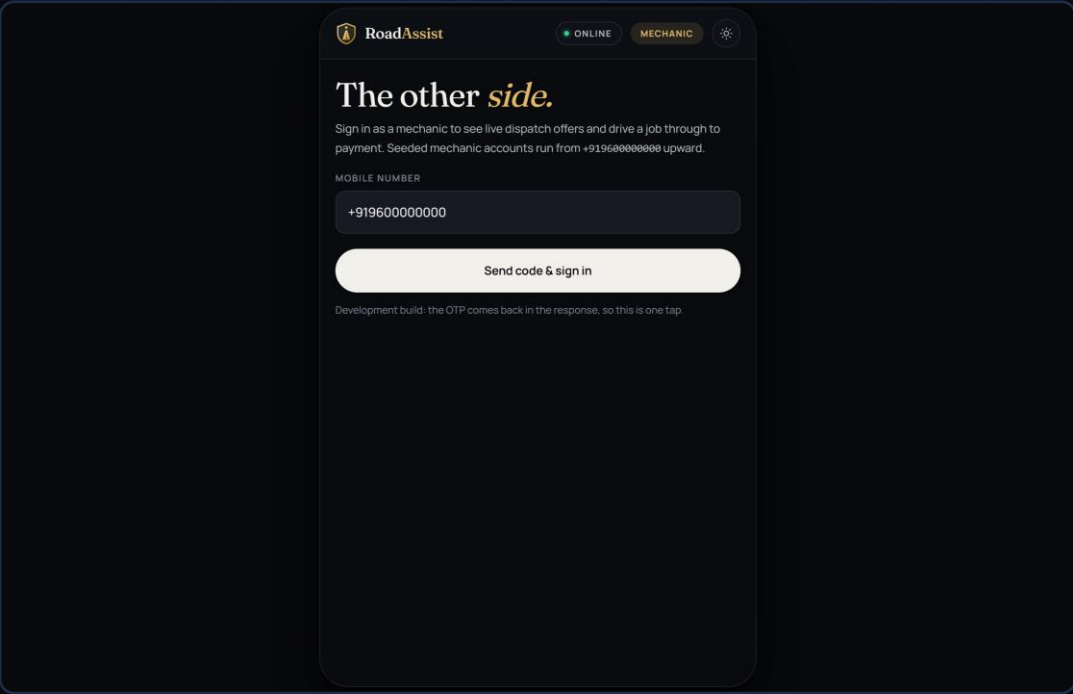
Dispatch



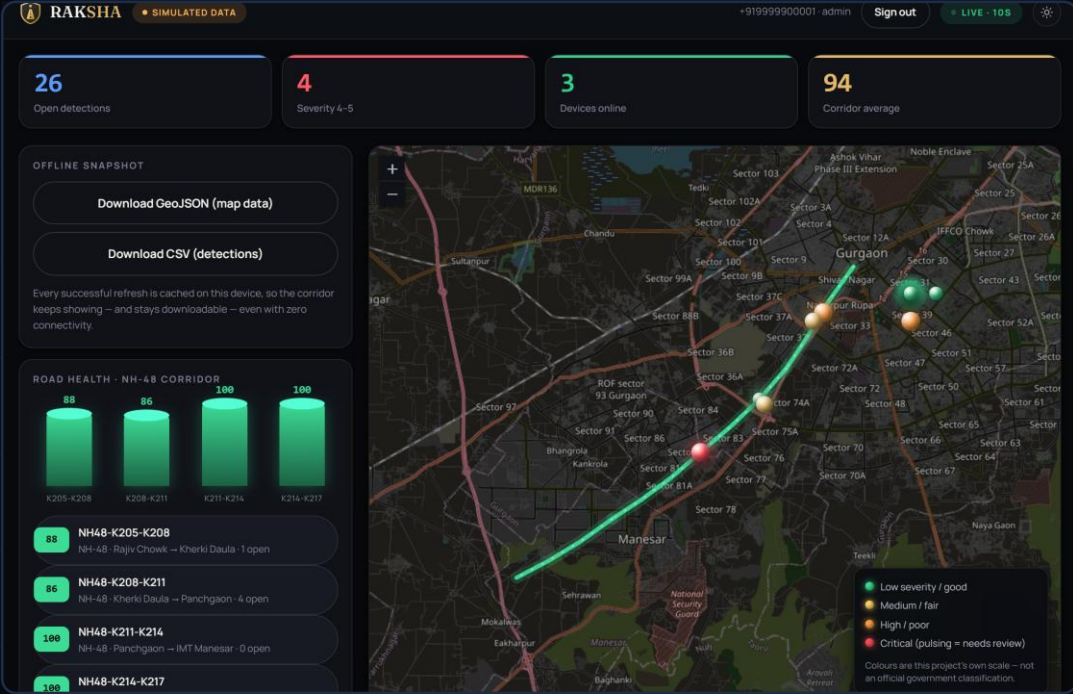
Activity

Captured by scripts/capture-screens.mjs against the live API — regenerable in ninety seconds, so they cannot drift from the product.

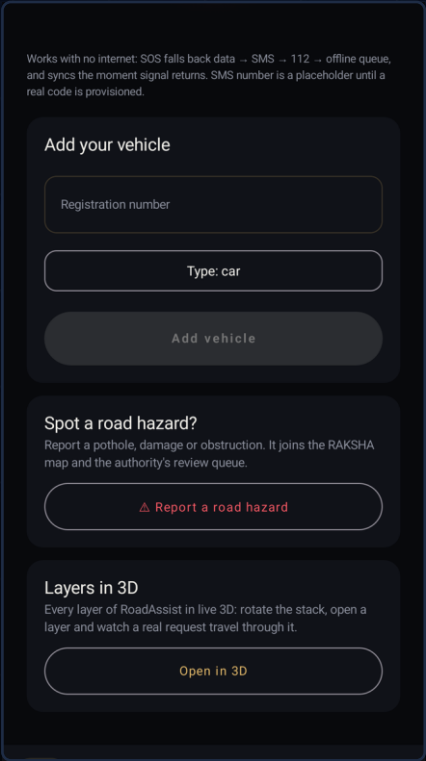
Mechanic console and authority dashboard



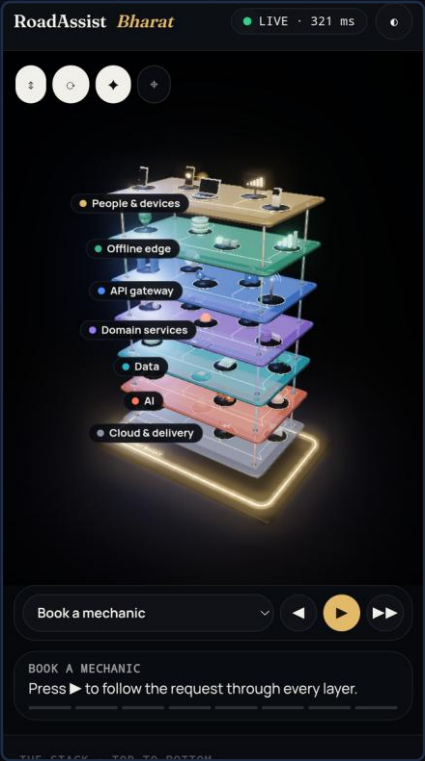
Mechanic console · dispatch inbox, live badge, job lifecycle
Mechanic: availability · incoming request · accept · decline · en route · arrived · in service · complete — every transition persisted and pushed to the customer.



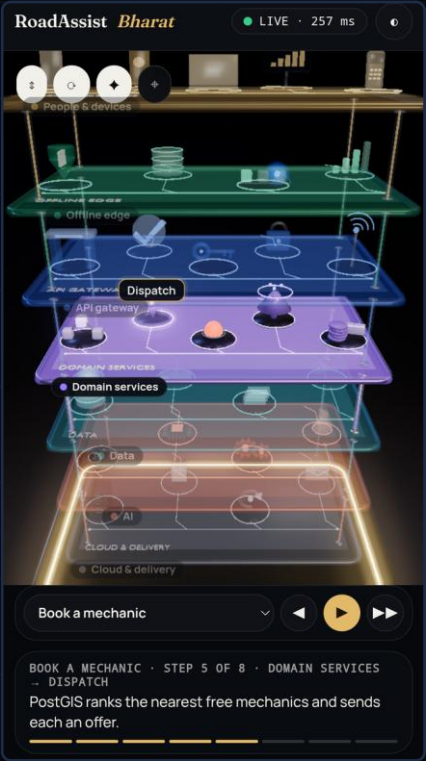
The Android app, on the emulator



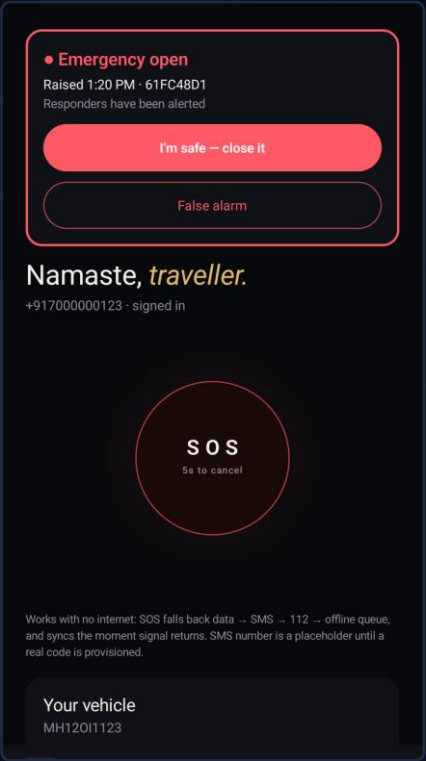
Home · the new Layers in 3D card



Layers in 3D · live, in a WebView



A journey playing · step 5 of 8



An open emergency, found after a restart

115 Android unit tests · 0 lint errors · 8 languages · the emergency card is shown against a local test server

Three states, three honest answers

ONLINE

- Everything works
- SOS reaches dispatch
- Live tracking + ETA
- Cloud diagnosis

LIMITED

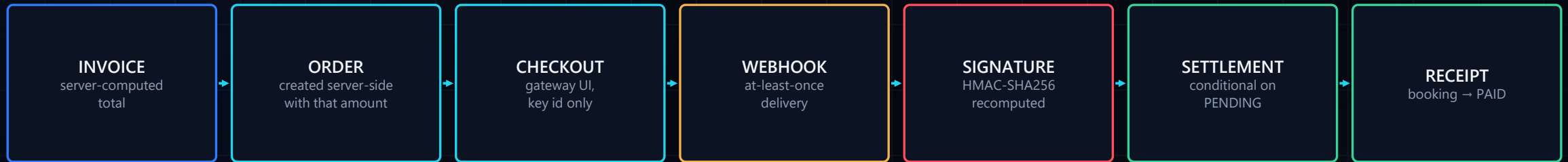
- Requests still attempted
- Anything that fails is STORED
- Slow or flaky path
- User is told it is weak

OFF-GRID

- SOS stored on device
- GPS still works
- Local diagnosis
- Cached maps · no live anything

The indicator names the state in WORDS as well as colour — status is never carried by tint alone.
Under every state the primary action stays the same and stays one tap away: SOS.

The client never decides that money arrived



AMOUNT

Never taken from the request — it is the invoice total, and the webhook refuses a mismatch

TWO PATHS

Browser callback AND webhook. The webhook is the one that matters: a customer can pay and close the tab

IDEMPOTENT

Delivery is at-least-once, so a replayed webhook settles nothing twice

REFUSES TO BOOT

Production will not start on a real gateway with no webhook secret

CHECKED, NOT COUNTED

22 checks against a local stub of Razorpay's API, no account: forged signature, replay, wrong amount — outside the 899

SEVEN LAYERS, AND 74 ATTACKS THAT FAIL

Security

The suite passes when the platform refuses — “we tried and could not get in” is a stronger statement than “the code looks right”.

AUTHENTICATION

OTP → JWT, rotating refresh
with reuse detection

RBAC

role gates on every
operator surface

RESOURCE OWNERSHIP

checked at the resource,
not just the route

RATE LIMITING

per principal — and never
on SOS escalation

INPUT VALIDATION

zod at every boundary,
parameterised SQL

PAYMENT VERIFICATION

server-side amount,
signed webhook

AUDIT LOG

hash-chained, append-only,
verified live

74 / 74

attacks refused

Sensitive data: medical break-glass needs a role, a LIVE incident and a written reason, and writes a tamper-evident row.

Logs redact credentials, OTP codes, phone numbers, medical fields and coordinates — and that redaction is unit-tested.

Three real vulnerabilities were found by this suite during the audit and fixed: missing request ids on 401s, a 500 on malformed JSON, an unsigned SMS intake that did not say so.

EACH NEW TEST FAILS ON THE OLD CODE AND PASSES ON THE NEW

Found in the recheck, and proved

WHAT WAS WRONG	BEFORE	AFTER
Twelve simultaneous code requests for one number	12 codes sent	5 · the cap holds
One sign-in code redeemed eight times at once	4 sessions	1 session
A customer marks their own job done	mechanic's public count 23 → 24	unchanged
Email sign-in answered slower for real accounts	revealed who is registered	same answer for all
Code-guessing alerts on protected accounts	could be crowded out	always sent
App restarted mid-emergency	could not close it	card: I'm safe · False alarm

Also fixed: a map tile retried once on a network blip, a 10 s deadline on the email service, a lock on the demo seed, a map that no longer asks for a request it knows will be refused, and eight fixes in the 3D page from a code review.

PostgreSQL 16 + PostGIS

Chosen for one query, and kept for four more reasons.

56

tables

62

foreign keys

138

indexes

5

GiST spatial

5

check constraints

5

migrations

WHY POSTGIS

“nearest available mechanic to this point” is ONE indexed query with ST_DWithin doing true metres-on-a-sphere. Without it: fetch every mechanic and compute haversine in app code.

WHY RELATIONAL

SELECT ... FOR UPDATE is what stops two mechanics taking one job. Transactions are the feature, not a formality.

CONSTRAINTS

client_incident_id UNIQUE is the entire duplicate-emergency defence. Money is integer paise; ratings are CHECK 1–5.

AUDITABILITY

audit_log is hash-chained and Postgres RULES block UPDATE and DELETE — an admin cannot quietly edit it.

VERIFIED

Migrations run from an EMPTY database in this audit, then the full suite. Backup + restore rehearsed: chain intact across 639 entries.

Where we sit in the cloud

We are a consumer of IaaS and PaaS, and a provider of SaaS to three user classes.

SaaS

We PROVIDE this: citizen app,
mechanic console, authority dashboard

IMPLEMENTED

PaaS

We CONSUME managed Postgres (Neon)
+ a container platform (Render)

IMPLEMENTED

IaaS

Compute, network, storage.
Compose is the local stand-in

PARTIAL

CHARACTERISTICS WE ACTUALLY EXHIBIT

Broad network access — one API serves a browser, an installed PWA, an Android WebView and a feature phone over SMS.
Resource pooling — dispatch pools providers and allocates on demand.

DEPLOYMENT MODEL

Public cloud. Demo: Render + Neon in Singapore — the free tiers offer no India region. Production target: an Indian region, because “No PII leaves India” — logs, backups and crash reports included.

MODULE 2

Enabling technologies

Five topics. Four are genuinely built; one is partial and says so.

Data centre

Compose stack: PostGIS + Redis + Redpanda on a private network; DB port NOT published in production

PARTIAL

Virtualization

Multi-stage container, non-root (uid 1000), tini as PID 1, healthcheck. Built and run — 321 MB.

IMPLEMENTED

Web technology

REST over HTTPS with a uniform envelope, server-sent events for real-time, PWA with a service worker

IMPLEMENTED

Multitenancy

Row-scoped shared schema. 12 cross-tenant attacks fired; all 12 refused.

IMPLEMENTED

Service technology

Versioned /v1, 69 routes (66 under /v1), zod validation, stable error codes, idempotency keys

IMPLEMENTED

MODULE 3

Infrastructure mechanisms

Virtual server	Container image; runs anywhere Docker runs. Full test suite passes AGAINST the image.	IMPLEMENTED
Cloud storage device	Postgres volume, upload volume, and client-side IndexedDB with AES-GCM-256 at rest	IMPLEMENTED
Cloud usage monitor	Structured JSON logs, per-request correlation id, operation duration/result, /v1/ops/overview with live counts	PARTIAL
Ready-made environment	compose → migrate → seed → start. Reference-only seed for production verified: 0 demo rows.	IMPLEMENTED
Network perimeter	DB has no published port; API on loopback behind a proxy; CORS allowlist enforced in production	PARTIAL
Resource replication	Stateless API is the precondition and is met. Nothing is replicated.	TARGET

MODULE 4

The eight fundamental architectures

Two are built. Six are design. I will not show you an animation and call it an autoscaler.

Workload distribution

The dispatch engine distributes jobs across a provider pool by computed score

IMPLEMENTED

Resource pooling

Providers are a pool; state is derived and busy members excluded

IMPLEMENTED

Dynamic scalability

HPA on queue depth. BUILT: the readiness gate — /health returns 503 when the DB is down

TARGET

Elastic resource capacity

Cluster autoscaler under the pods

TARGET

Service load balancing

LB with health-checked rotation

TARGET

Cloud bursting

Modelled evening scenario — labelled as modelled

TARGET

Elastic disk provisioning

Storage allocated as data arrives

TARGET

Redundant storage

Replica with failover. Backup procedure documented and rehearsed.

TARGET

Three things block replication today, and each is documented where it is defined: the SSE registry, the rate limiter and the offer sweeper are in-process.

The fix for each is already designed — Redis, and the outbox → Redpanda bus already in the architecture.

Static vs dynamic scheduling

STATIC

Decided before run time, on a fixed schedule.

- Offer sweeper — every 10 s (OFFER_SWEEP_SECONDS)
- Client booking poll — 6 s, or 60 s while streaming
- SSE heartbeat — every 25 s
- Retention purge — 24 h after a sync

DYNAMIC

Decided at run time, from live state.

- Dispatch: pool queried at request time
- Score from live distance, rating, workload
- Wave offered; timeout re-schedules to the next wave
- A decline escalates immediately
- Assignment settled under a row lock — exactly one

RoadAssist's dispatch engine IS dynamic scheduling — it is the clearest Module 4 mechanism the project actually implements.

TESTING

899 assertions, all executed

Against a real PostgreSQL + PostGIS and a real Chrome. 22 Razorpay stub checks sit outside this total.

Unit	225	state machines, rules, redaction, SMS coordinates, i18n segment budget
------	-----	--

End-to-end	194	the whole API journey against real Postgres
------------	-----	---

Concurrency + real-time	77	paces a sequential suite structurally cannot make
-------------------------	----	---

Security	74	attacks that must FAIL
----------	----	------------------------

Gateway security	27	webhook signatures, append-only rules, OTP ceilings
------------------	----	---

Browser / offline	163	what only a browser can prove
-------------------	-----	-------------------------------

899 passed · 0 failed · 0 skipped · 0 server errors across the sweep

Every failure has an answer

DETECT → DEGRADE → PRESERVE DATA → RECOVER. Twenty rows in the failure matrix; seven of them here.

FAILURE	DETECTED BY	USER EXPERIENCE	RECOVERY
Network lost	connectivity manager	off-grid mode; SOS stored locally	auto on reconnect
Backend unreachable	probe fails, radio up	LIMITED, then OFF-GRID	auto
Database down	/health → 503	safe 500 with a request id	auto, no restart — tested
GPS denied	geolocation error	“Unknown — denied”, never a fake fix	next fix
AI endpoint down	timeout	rules engine — nothing visible	auto
Payment fails	gateway error / bad signature	invoice stays unpaid	customer retries
Provider times out	expires_at passes	sweeper escalates to next wave	ladder continues

Each row names the suite that proves it. Database loss and backup/restore are now automated CI steps, not prose.

IMPLEMENTATION STATUS

What is built, what is partial, what is design

Marked from the code, not from intent.

AI diagnosis rules engine + trained YOLO11 for road damage	PARTIAL	Real-time SSE, measured 65 ms	IMPLEMENTED
SOS — online e2e + concurrency suites	IMPLEMENTED	Payment mock on the demo — never a live account	IMPLEMENTED
SOS — offline browser suite drives the whole scenario	IMPLEMENTED	Fleet schema + roles exist; no dedicated UI	PARTIAL
Offline storage IndexedDB, AES-GCM-256, survives restart	IMPLEMENTED	Analytics ops overview with live counts; no BI layer	PARTIAL
Store-and-forward sync idempotent; duplicate proven impossible	IMPLEMENTED	Cloud autoscaling designed; no cluster	TARGET
Dispatch ladder, timeout, busy-exclusion, race-safe	IMPLEMENTED	Replication / load balancing stateless API is the precondition, and is met	TARGET

GREEN = code exists, database persists it, UI consumes it, AND a test exercises it. AMBER = some of that. BLUE = architecture only.

Limitations

In order of importance. Every one of these is also written in the code or the docs.

- 1 The demo is one free-tier instance in Singapore (Render + Neon) — no India region, no cluster.
- 2 Single instance only: SSE registry, rate limiter and offer sweeper are in-process.
- 3 The ERSS 112 handoff is a stub, and the API response says so. Emergency isolation (ADR-0005) is a design, not a deployment.
- 4 Payment checks run only against a local stub of Razorpay's API, outside the 899 — never a real account.
- 5 The diagnosis “AI” is a deterministic rules engine. Labelled as such everywhere.
- 6 No load test, no external penetration test, no coverage on the HTTP layer (suites run out-of-process).
- 7 Device encryption protects a storage dump, not script on the same origin — and the UI says exactly that.
- 8 GPS cold start can take minutes and may never fix indoors. The app shows “Unknown” rather than a guess.
- 9 No historical volume yet for predictive maintenance — the dead-zone heuristic is labelled heuristic v1.

FUTURE — LABELLED AS SUCH

Roadmap

NEXT

- Redis-backed rate limiting
- Outbox → event bus for SSE fan-out
- WAL archiving (RPO under a day)
- Object storage for photos

THEN

- Multi-region deployment
- Database read replicas
- External APM + error reporting
- Automated backup schedule

LATER

- Predictive maintenance at volume
- Fleet management UI
- Satellite emergency communication
- Mesh networking between devices
- Government emergency-network integration

Satellite, mesh and government-network integration are FUTURE and are labelled that way everywhere in the product — not implemented, not claimed.

Where the money would come from

Potential, not current. No commercial operation exists — this is a university project.

B2C

Subscription for individual drivers; pay-per-incident for non-members

B2B

Fleet operators — uptime is their cost centre; per-vehicle per-month

B2B2C

Insurers and OEMs bundling roadside cover into an existing policy

B2G

Highway authorities — RAKSHA road-health data and incident analytics

The unit economics that make it plausible: the platform runs on ZERO paid third-party accounts by default.
Maps are keyless OpenStreetMap, AI is a local rules engine, and every vendor sits behind an adapter with a working local implementation.

Current status: no revenue, no customers, no commercial deployment. Stated so the model is read as a plan, not a claim.

What a competitor would have to build

Not “we are better than X” — we have not benchmarked anyone. This is what is genuinely hard here.

Network resilience

An emergency workflow that runs entirely on the device, with an encrypted journal and idempotent forward-sync. This is the moat.

Honest degradation

The product never claims something reached the cloud when it did not. That is a design discipline, not a feature you can add later.

Race-safe dispatch

Row-locked assignment, wave ladder, timeout escalation, derived provider state — proven with ten simultaneous accepts.

Explainable diagnosis

The same rule table on device and cloud, with a CI test that fails the build if they diverge.

Auditability

Hash-chained append-only log, verified live in the operations view.

Live demonstration

PART ONE · THE FULL JOURNEY

1. Login
2. Vehicle
3. Breakdown + AI diagnosis
4. Request assistance
5. Dispatch — ranked offers
6. Mechanic accepts
7. Live status, no refresh
8. Service → invoice → payment
9. Review

PART TWO · THE ONE NOBODY ELSE HAS

1. Network OFF — the UI changes immediately
2. Local diagnosis, labelled
3. Hold SOS → off-grid incident, real GPS
4. "Nothing has been transmitted"
5. CLOSE THE TAB. REOPEN IT.
6. The incident is still there
7. Network ON → automatic sync
8. "SOS synchronized"
9. Replay the reference → duplicate refused

If any step fails, the failure IS the demonstration — nothing here fakes a success.

RoadAssist Bharat

From breakdown to resolution — even when connectivity fails.

AI

explainable
rules engine

CLOUD

consumed and
provided

OFFLINE

the whole emergency
path on-device

REAL-TIME

65 ms, and never
the only path

SECURITY

87 attacks,
all refused

DISPATCH

dynamic, race-safe
scheduling

“RoadAssist doesn't stop when the network stops.”

899 assertions · 6 suites · 0 failures · 56 tables · 69 routes · 11 ADRs · every claim on these slides is testable